मोतीलाल नेहरू राष्ट्रीय प्रौद्योगिकी संस्थान इलाहाबाद
प्रयागराज–211004 भारत

Motilal Nehru National Institute of Technology Allahabad
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Computer Science and Engineering Department
End Semester (Odd) Examination 2025-26

Programme Name: B.Tech	**Semester: V**
Course Code: CSN15403	**Course Name: Cryptography and Network Security**
	Student Reg. No.:
Duration: 150 Minutes	**Max. Marks: 50**

Instructions: (Related to Questions)

1. *Precise answer with proper justification will be considered to get complete marks.*
2. *Attempt all questions and in the same order.*

			Marks
Q1	**a.**	Data Encryption Standard is a symmetric key block cipher considered under Feistel Ciphers. Data Encryption Standard and Double Data Encryption Standard were not successful. Discuss the issues, attacks, in DES and Double DES and justify how these limitations were addressed.	**4**
	b	Assume Captain America wants to send a message to Thor using RSA algorithm such that Thanos should not get the message. The message is 63, modulus is 77. Choose encryption exponent as 13. Find decryption exponent and cipher text using fast exponentiation method.	**6**
Q 2	a	The ElGamal signature scheme involves the use of the private key for digital signature generation and the public key for digital signature verification. Let us consider the prime field GF(19). Answer the following questions. i. Find the primitive roots. ii. Prove the correctness of the algorithm with respect to verification and signatures. iii. Let us assume the random integer is r=5, one of the primitive root e1=10 and	**9**

		sender's private key d=16. Find the value of e2, S1 and S2 for Message M=14 and verify the signature of the algorithm.	
Q3	**a**	Secret sharing scheme in cryptography aims to distribute and later recover secret S among n parties. Secret S is distributed in form of shares which are generated from secret. Without cooperation of k no of parties, the secret cannot be reconstructed from shares directly. The shares for n=3 number of parties are generated taking modulus 3, 5 and 7 they are respectively 2, 3 and 2. Recover the secret S from the above information.	**5**
Q4	**a**	Finding an algorithm to correctly and efficiently test a very large integer and output a prime or a composite has always been a challenge in number theory, and consequently in cryptography. However, we can make the probability of error so small that it is almost certain that the algorithm has returned a correct answer. Justify the above statement by elaborating on Miller Rabin Test for Primality with algorithm and verify 1019 is a prime or not?	**7**
Q5	**a**	SHA-512 is a member of the SHA-2 family and operates on 1024-bit message blocks using 64-bit words. It relies on a sequence of logical functions and modular arithmetic to achieve diffusion and non-linearity. Explain SHA-512 design and which includes message preparation, word expansion, compression function and structure of each round.	**7**
Q6	**a**	Assume that generator is 7 and prime is 23, and say Srikanth chooses his secret value 3 and JK Talpade chooses his secret value 6. Using Diffie-Hellman method show how they both create session key for secret communication. How Rukma can intercept the secret communication. Elaborate with a scenario	**6**
	b.	Write Short Notes on the following i. Hill Cipher ii. Steganography iii. Euler's Theorem	**6**